Software Engineering

Assignment-12

TYPE OF QUESTION: MCQ/MSQ

Number of questions: **10** **Total mark: 10 X 1 = 10**

For each of the following questions one or more of the given options are correct. Choose the correct options.

QUESTION 1:

Which of the following statements regarding cyclomatic complexity are true?

a. Higher cyclomatic complexity for a code indicates higher code quality
b. Higher cyclomatic complexity may indicate more potential errors in code
c. Cyclomatic complexity correlates well with execution speed directly
d. Programs with higher complexity may require more effort to understand
e. Cyclomatic complexity can be related to the effort required for debugging

Correct Answer: b. Higher cyclomatic complexity may indicate more potential errors in code

d. Programs with higher complexity may require more effort to understand

e. Cyclomatic complexity can be related to the effort required for debugging

Detailed Solution:

Higher cyclomatic complexity usually indicates more complex code, which may increase error probability, understanding difficulty, and debugging effort

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QUESTION 2:

Which of the following are true regarding the coverage strength of various white-box testing techniques?

a. Statement coverage is among the weakest criteria shown
b. All path coverage is among the strongest criteria shown
c. Branch coverage guarantees multiple-condition coverage
d. MC/DC is stronger than branch and condition coverage
e. Condition coverage alone guarantees branch coverage

Correct Answer: a. Statement coverage is among the weakest criteria shown, **b**. All path coverage is among the strongest criteria shown, **d.** MC/DC is stronger that branch and condition coverage

Detailed Solution:

Statement coverage is relatively weak, while all path coverage is very strong. MC/DC is stronger than branch and condition coverage because it checks independent effect of each condition.

QUESTION 3:

Which of the following statements about basis path coverage are true?

a. It is also known as all path coverage
b. It is stronger than statement coverage
c. It guarantees exhaustive testing of all possible paths
d. It is related to cyclomatic complexity
e. It is weaker than branch coverage

Correct Answer: b. It is stronger than statement coverage, **d.** It is related to cyclomatic complexity

Detailed Solution:

Basis path coverage tests independent execution paths and is stronger than statement coverage. It is directly related to cyclomatic complexity.

QUESTION 4:

Which of the following statements regarding data flow testing are true?

a. It focuses on the definition and use of variables in a program
b. It is a fault-based testing technique
c. It ignores control flow in the program completely
d. It helps detect anomalies such as variables being used before initialization
e. It is based on tracking the life cycle of data objects

Correct Answer: a. It focuses on the definition and use of variables in a program
d. It helps detect anomalies such as variables being used before initialization
e. It is based on tracking the life cycle of data objects

Detailed Solution:

Data flow testing analyzes variable definitions and uses to detect anomalies like using variables before initialization by tracking the data life cycle.

QUESTION 5:

Which of the following statements regarding mutation testing are true?

a. Equivalent mutants indicate that additional or improved test cases may be needed
b. Mutants are created by introducing small changes into the program
c. A mutant is said to be “killed” if a test case fails due to the change
d. Mutation testing is primarily a black-box testing technique
e. Mutation testing helps evaluate the effectiveness of a test suite

Correct Answer: b. Mutants are created by introducing small changes into the program
c. A mutant is said to be “killed” if a test case fails due to the change
e. Mutation testing helps evaluate the effectiveness of a test suite

Detailed Solution:

Mutation testing creates small code changes called mutants. A mutant is “killed” when test cases detect the change, helping evaluate test suite effectiveness.

QUESTION 6:

Which of the following are objectives of mutation testing?

a. Assess the fault-detection capability of test cases
b. Improve the quality of the test suite
c. Guarantee absence of all software defects
d. Identify weak test cases and eliminate them from the test suite
e. Identify duplicate and equivalent test cases and optimize the test suite

Correct Answer: a. Asses the fault detection capability of test cases, **b.** Improve the quality of the test suite

Detailed Solution:

Mutation testing evaluates how effectively test cases detect faults and helps improve the quality of the test suite.

QUESTION 7:

In the context of mutation testing, suppose in a C program, the statement **if(a > b) c=0;** is replaced by **if(a>=b-1) c=0;** What kind of mutant has most likely been created?

a. Trivial mutant
b. Stillborn mutant
c. Higher-order mutant
d. Equivalent mutant
e. Syntax mutant

Correct Answer: d. Equivalent mutant

Detailed Solution: The modified condition behaves similarly to the original for many integer values, so the mutant may produce the same output as the original program.

QUESTION 8:

What would be the Cyclomatic complexity measure of the following program?

```
int check(int a, int b, int c) {
    int result;

    if (a < b)
        if (a < c)
            result = a;
        else
            result = c;
    else
        if (b < c)
            result = b;
        else
            result = c;

    return result;
}
```

a. 2
b. 3
c. 4
d. 5
e. 6

Correct Answer: c. 4

Detailed Solution:

There are 3 if conditions, so complexity = $3 + 1 = 4$.

QUESTION 9:

Which of the following are major challenges associated with mutation testing?

a. Mutation testing improves the fault-detection capability of test cases
b. Generating and managing mutants needs to be done manually, thereby resulting in a dramatic increase in testing costs
c. Large numbers of mutants may need to be executed
d. Equivalent mutants are difficult to identify automatically
e. Mutation testing is not as effective as other testing techniques

Correct Answer: c. Large numbers of mutants may need to be executed, **d.** Equivalent mutants are difficult to identify automatically

Detailed Solution:

Mutation testing may require executing many mutants, and equivalent mutants are difficult to detect automatically, making the process expensive and complex.

QUESTION 10:

At least how many test cases are required to achieve MC/DC coverage of the following code segment:

```
if ((p > 10) && (q < 50))   y = y + 1;
```

a. 1
b. 2
c. 3
d. 4
e. 6

Correct Answer: c. 3

Detailed Solution:

MC/DC requires each condition to independently affect the decision outcome. For two conditions, the minimum required test cases are 3.
